

Homework 2: matrix factorizations and principal component analysis

Submission

Due: 09/21/2026

Submit the following three files in your directory:

1. `lastname_firstname_hw02.ipynb`: one executed Jupyter notebook containing all code and markdown text;
2. `lastname_firstname_hw02.pdf`: a PDF compiled or exported from the executed notebook; and
3. `requirements.txt`: a short list of the packages needed to run the notebook, with version numbers (for example, `numpy==2.1.0`).

Your notebook must run from top to bottom in a fresh session without manual edits. Use NumPy and SciPy for the required matrix factorizations rather than implementing the factorization algorithms from scratch. Report requested numerical results with at least three significant digits.

Question 1: Cholesky factorization and repeated solves (50 points)

Use the following code to generate a symmetric positive-definite matrix and 40 right-hand sides:

```
import numpy as np

rng = np.random.default_rng(123)
m = 1000
G = rng.normal(size=(m, m))
A = G @ G.T / m
B = rng.normal(size=(m, 40))
```

1. Compute the Cholesky factorization $A = LL^\top$. Report

$$\frac{\|A - LL^\top\|_F}{\|A\|_F}$$

and the smallest eigenvalue of A . Explain how these values support the validity of the factorization.

2. Solve $AX = B$ by first solving $LY = B$ and then $L^\top X = Y$. You may use `scipy.linalg.solve_triangular`. Report the relative residual

$$\frac{\|AX - B\|_F}{\|B\|_F}$$

and the relative difference of such computed X from `np.linalg.solve(A, B)`; notice that the denominator should be the F-norm of the returned result from `np.linalg.solve(A, B)`.

3. Compare the wall-clock time of:

- 40 separate calls to `np.linalg.solve`, one for each column of B ; and
- one Cholesky factorization followed by 40 triangular solves for all 40 columns.

Include the factorization time in the Cholesky workflow. Repeat each timing three times and report the median. Briefly explain why reusing a factorization is valuable when A remains fixed.

Question 2: SVD and PCA (50 points)

Use the following code to generate eight observed variables from two latent factors:

```
rng = np.random.default_rng(456)
n = 600
latent = rng.normal(size=(n, 2))
loadings = np.array([
    [1.0, 0.9, 0.8, 0.7, 0.0, 0.0, 0.2, 0.1],
    [0.0, 0.0, 0.1, 0.2, 1.0, 0.9, 0.8, 0.7],
])
X = latent @ loadings + 0.15 * rng.normal(size=(n, 8))
group = (latent[:, 0] + latent[:, 1] > 0).astype(int)
```

1. Center each column of X and compute its SVD. Use the singular values to calculate the explained-variance ratio of every principal component. Report the ratios and the smallest number of components needed to explain at least 90% of the total variance.
2. Construct the rank-2 approximation X_2 of the centered data. Report its relative reconstruction error

$$\frac{\|X_{\text{centered}} - X_2\|_F}{\|X_{\text{centered}}\|_F}.$$

Verify numerically that this equals

$$\sqrt{\frac{\sum_{j>2} \sigma_j^2}{\sum_j \sigma_j^2}}.$$

3. Produce a scatterplot of the first two principal-component scores, colored by **group**.

Label both axes clearly. Interpret the effective dimension of the data and the visible structure in the score plot.

Grading guide

Criterion	Points
Correct implementation and numerical results	50
Sound analysis and interpretation	25
Clear figures, tables, and concise communication	15
Reproducibility and complete submission	10
Total	100

Minor differences caused by software versions and timing variability are acceptable. Numerical results without written interpretation will receive limited credit.